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The equation $x^4 + 2x + 4y^3 + y - 1 = 0$ has no integer solutions

Size $h = 55$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + 2x + 4y^3 + y - 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), with $(d_1, d_2) = (-6, 1)$, and the fact that an odd prime p with $(-d_k/p) = -1$ divides $U_k^2 + d_k V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm forms $a^2 + db^2$

Throughout this section, d is a squarefree integer with $d \neq 0, -1$; we apply the results with $d \in \{-6, 1\}$. For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(-d/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + db^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv -d(bc)^2 \equiv -d \pmod{p}$, so $-d$ is a square modulo p . As $(-d/p) \neq 0$, it is a non-zero square, so $(-d/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 + db^2 \neq 0$, and $v_p(a^2 + db^2)$ is even for every odd prime p with $(-d/p) = -1$.*

Proof. If $d > 0$, then clearly $a^2 + db^2 > 0$. If $d < 0$, then $-d > 1$ is squarefree, hence not a square, so $a^2 + db^2 \neq 0$. Let p be an odd prime with $(-d/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + db'^2$ by Lemma 2. Since $a^2 + db^2 = p^{2k}(a'^2 + db'^2)$, we get $v_p(a^2 + db^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{4|d|}$, or, if $d < 0$, with $n' \equiv -n \pmod{4|d|}$. Then $(-d/n') = (-d/n)$.*

Proof. Since d is squarefree, we can write $-d = \varepsilon 2^e q$ with $\varepsilon \in \{1, -1\}$, $e \in \{0, 1\}$ and q odd and positive. By multiplicativity and the reciprocity laws,

$$\left(\frac{-d}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since $4|d|$ is divisible by $4q$, and by $8q$ if $e = 1$, the claim for $n' \equiv n$ follows. If $d < 0$ and $n' \equiv -n \pmod{4|d|}$, then, as $\varepsilon = 1$, the factor (ε/n) does not change, $(2/n)$ does not change because $n'^2 \equiv n^2 \pmod{16}$, and each of $(-1)^{(n-1)(q-1)/4}$ and (n/q) is multiplied by $(-1)^{(q-1)/2}$, because $(n' - 1)/2 \equiv (n - 1)/2 + 1 \pmod{2}$ and $(n'/q) = (-1/q)(n/q) = (-1)^{(q-1)/2}(n/q)$. $\square$

Lemma 5. *Let n be a non-zero integer, and let s, u and m be positive integers such that every prime factor of s divides $2|d|$, both $2s$ and $4|d|s$ divide m , u is odd, $(-d/u) = -1$, and $n \equiv su \pmod{m}$. If $d > 0$, suppose also that $n > 0$. Then there is an odd prime p with $(-d/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by $4|d|$, n' is odd and $n' \equiv u \pmod{4|d|}$. If $d > 0$, then $n' > 0$, and if $d < 0$, then $|n'| \equiv \pm u \pmod{4|d|}$; in both cases Lemma 4 gives $(-d/|n'|) = (-d/u) = -1$. By multiplicativity, the product of $(-d/p)^{v_p(n')}$ over the prime factors p of $|n'|$ is -1 , so some prime factor p of n' satisfies $(-d/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 2d$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + 2x + 4y^3 + y - 1$, so that (1) is the equation $F(x, y) = 0$. For the first identity, $d_1 = -6$, and we define

$$\begin{aligned} H_1 &= 3x^2 + 4xy - 2x + 2y^2 + 2y + 1, \\ T_1 &= 16x^5 - 8x^4y + 80x^4 - 96x^3y - 96x^3 + 80x^2y^2 + 16x^2y \\ &\quad + 120x^2 + 160xy^2 - 16xy + 16x + 24y^2 - 8y + 2, \\ U_1 &= 2x^4 + 4x^3 + 2x^2 - 134x + 76, \\ V_1 &= -2x^3 + 55x - 31, \\ G_1 &= -4x^4 + 32x^3 + 40x^2y + 208x^2 + 80xy + 96x + 12y + 8, \\ A_1 &= -348482x^2 - 198358x + 9477449, \\ B_1 &= -348482x^3 - 895322x^2 - 851004x + 22884950, \end{aligned}$$

for the second identity, $d_2 = 1$, and we define

$$\begin{aligned} H_2 &= x^2 - 2y + 2, \\ T_2 &= x^4 + 2x^2y + 4x^2 + 4y^2 + 4y + 5, \\ U_2 &= x^3 + 2x + 2, \\ V_2 &= -2x^2 + x - 2, \\ G_2 &= -2, \\ A_2 &= -10x + 17, \\ B_2 &= -5x^2 + 6x - 2, \end{aligned}$$

and, for the signs, we also use

$$\begin{aligned} Q_2 &= 4x^4 + 8x^2y + 16x^2 + 16y^2 + 16y + 20, \\ M_2 &= 4x^6 + 32x^4 + 52x^2 + 16x + 32, \\ E_2 &= 192x^4 + 768x^2 + 1024. \end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 + 6V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 - V_2^2 = G_2F, \quad (3)$$

$$A_1U_1 + B_1V_1 = 2 \cdot 3 \cdot 7 \cdot 233 \cdot 1109, \quad (4)$$

$$A_2U_2 + B_2V_2 = 2 \cdot 19, \quad (5)$$

$$-8F = Q_2H_2 - M_2, \quad (6)$$

$$64Q_2 = (8x^2 + 32y + 16)^2 + E_2. \quad (7)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) and (3) give

$$H_kT_k = U_k^2 + d_kV_k^2 \quad (k = 1, 2), \quad (8)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 648^2 residue pairs modulo 648, we find that (1) has exactly 648 solutions modulo 648, and that at each of them $H_1 \bmod 648$ lies in the set $\mathcal{R}_1$ listed in Table 1 or $H_2 \bmod 648$ lies in the set $\mathcal{R}_2$ listed in Table 1. For $k = 1$, each $r \in \mathcal{R}_1$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(6/u) = -1$; here every prime factor of each $s \in \{1, 9, 27\}$ divides 12, and $2s$ and $24s$ divide 648 for each such s . For $k = 2$, each $r \in \mathcal{R}_2$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-1/u) = -1$; here for $s = 1$, $2s = 2$ and $4s = 4$ divide 648.

Step 2: signs. *The value of H_1 .* By (4), U_1 and V_1 are not both zero, so $U_1^2 - 6V_1^2 \neq 0$ by Lemma 3 with $d = -6$, and (8) gives $H_1 \neq 0$.

The sign of H_2 . Since $F(x, y) = 0$, (6) gives $Q_2H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_2 > 0$ and $E_2 > 0$. Hence $M_2 > 0$ and $E_2 > 0$, so (7) gives $64Q_2 \geq E_2 > 0$, and $Q_2H_2 = M_2 > 0$ gives $H_2 > 0$.

Step 3: primes p with $(-d_k/p) = -1$. Let $k \in \{1, 2\}$, and let p be an odd prime with $(-d_k/p) = -1$, where $(d_1, d_2) = (-6, 1)$. We show that $v_p(H_k)$ is even. By (4)–(5), U_k and V_k are not both zero, so (8) and Lemma 3 show that $v_p(H_kT_k) = v_p(U_k^2 + d_kV_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 + d_kV_k^2$ by (8), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_kU_k + B_kV_k$. For $k = 1$, we have $R_1 = 2 \cdot 3 \cdot 7 \cdot 233 \cdot 1109$. As p is odd and $p \neq 3$ and $p \neq 1109$ because $(6/3) = 0$ and $(6/1109) = 1$, we get $p \in \{7, 233\}$. Checking the 7^2 residue pairs modulo 7 and the 233^2 residue pairs modulo 233, we find that F , H_1 and T_1 have no common zero modulo 7 and 233, a contradiction. For $k = 2$, we have $R_2 = 2 \cdot 19$. As p is odd, we get $p = 19$. Checking the 19^2 residue pairs modulo 19, we find that F , H_2 and T_2 have no common zero modulo 19, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_kT_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{648}$ with s and u as in Lemma 5 for $d = d_k$. By Step 2, $H_k \neq 0$, and $H_k > 0$ if $d_k > 0$. Hence Lemma 5 applies to $n = H_k$ and $d = d_k$, so there is an odd prime p with $(-d_k/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The sets $\mathcal{R}_k$ of residues modulo 648 in Step 1.

$\mathcal{R}_1$, the residues of H_1					
13	133	253	373	517	637
37	157	277	397	541	
61	181	301	421	$549 = 9 \cdot 61$	
85	$189 = 27 \cdot 7$	325	445	565	
109	205	$333 = 9 \cdot 37$	469	589	
$117 = 9 \cdot 13$	229	349	493	613	
$\mathcal{R}_2$, the residues of H_2					
7	119	223	335	439	551
23	127	239	343	455	559
31	143	247	359	463	575
47	151	263	367	479	583
55	167	271	383	487	599
71	175	287	391	503	607
79	191	295	407	511	623
95	199	311	415	527	631
103	215	319	431	535	647